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RESEARCH ARTICLE

Early Detection of Multiclass Skin Lesions Using Transfer Learning-Based IncepX-Ensemble Model

SUBHAJIT CHATTERJEE[1], JOON-MIN GIL[1], AND YUNG-CHEOL BYUN[2]
[1]Department of Computer Engineering, Jeju National University, Jeju 63243, South Korea
[2]Department of Computer Engineering, Institute of Information Science & Technology, Jeju National University, Jeju 63243, South Korea

Corresponding author: Yung-Cheol Byun (ycb@jejunu.ac.kr)

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ABSTRACT Skin lesion diagnosis in medical image analysis is still a difficult task. A frequent kind of cancer known as skin cancer affects people worldwide and can be fatal. As a result, early and accurate diagnosis is crucial for finding skin cancer patients. One of the most recent technologies for detecting skin cancer is dermatoscopy. For proper treatment of skin cancer, an early diagnosis is required. The early stages of skin lesions are identical, making manual diagnosis difficult. Therefore, creating computer-aided methods for classifying skin lesions will assist dermatologists in detecting skin lesions earlier and treating them more successfully. Current research indicates a significant potential for the classification of skin lesions using deep learning networks. However, issues including unbalanced datasets, poor contrast lesions, and the extraction of pointless or duplicate features still need to be resolved. This study aims to propose a transfer learning-based ensemble model for more accurate classification results. InceptionV3 and Xception are employed to build an ensemble model for classifying skin lesion images known as IncepX-Ensemble. A traditional data augmentation method was employed for the HAM10000 dataset to address the imbalanced data. This technique mitigates the class imbalance by incorporating data augmentation, enhancing model accuracy. At the experiment's outset, we utilized the original imbalanced data, and subsequently, balanced data were employed for the proposed model. Training and testing accuracy was achieved at 86% on the imbalanced dataset. A balanced data augmentation dataset yielded 98% training and 98% test accuracy rates. We evaluated the output of the proposed model against outputs from various transfer learning models using both the original and balanced datasets.

INDEX TERMS Deep learning, skin lesion classification, convolutional neural network, transfer learning, ensemble learning, image classification.

I. INTRODUCTION

The human skin, consisting of multiple layers, stands as the body's most crucial and sun-exposed organ. Enhancing the skin's role as a protective barrier involves employing fluids to prevent the degradation of lipids in the epidermis. Various factors, including fungal growth on the skin, imperceptible bacteria, bacterial alterations affecting skin texture, or changes in pigmentation, can give rise to skin symptoms [1].

The abnormal growth of skin cells is another name for skin cancer, and skin lesion cases have recently sharply increased worldwide [2]. Skin cancer is caused by exposure to UV light from the sun, not by a depletion in the ozone layer, which protects from those rays. Basal cell carcinoma, melanoma, squamous cell carcinoma, and actinic keratosis (solar keratosis) are some of the different types of skin cancer. Out of all the other types of skin cancer, melanoma is the most dangerous. Melanoma starts in the brown or black melanocyte cells [3], which resemble a mole. Melanoma cells penetrate deeper skin layers from the skin's surface. The cells (melanocytes) that create melanin, the pigment

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that gives your skin its color, and which are responsible for melanoma, the most serious form of skin cancer, develop. Melanoma has a 7% incidence rate, yet it accounts for 75% of fatal cancer cases [4]. The head, neck, face, and hands are just a few body places where skin cancer can develop due to increased exposure to the sun's harmful rays. Skin lesions must be identified and treated as soon as possible for the patient to survive. Given the resemblance between skin cancer, it can be challenging for dermatologists with experience to determine a skin cancer class. However, studies suggest that even skilled dermatologists can detect skin cancer with unaided eyes and visually examine it with an accuracy of about 75%-80% [5], [6]. Dermatologists often use the ABCD rule to diagnose skin cancer. Asymmetry, border, color, and diameter are the components of the ABCD rule [7]. Physicians identified the following features in skin lesions during the diagnosis. Dermatologists' diagnostic procedures are quite individualized. Numerous elements, including low picture resolutions, less expertise, and visual impairment, can lead to discrepancies in dermatologists' diagnoses and outcomes. These issues can have less effect if computer-based diagnosis systems are developed to enable such diagnostic applications [8]. Researchers may encounter misleading multiclass classification of skin cancer when the dataset is unbalanced. As a result, it's important to employ a data-balanced technique that will balance the classes, which is vital for the final classification. Due to the similarity of skin lesions and the unbalanced nature of the data, the multiclass classification problem is more difficult than the binary class classification problem. Because of the skin cancer differences in skin surfaces, identifying the exact place is difficult. As a result, even for seasoned dermatologists, manually identifying skin lesions via dermatoscopy takes time to determine and penetrate more error potential. Because of this, researchers emphasize various computer-aided diagnostic (CAD) methods based on machine learning. Dermatologists use computerized dermatoscopy examination to transmit images of skin lesion samples taken from the human body to the computer and assess the images to determine whether the lesion is worrisome. Improvements in machine learning techniques are beneficial for disease diagnosis and image analysis in medicine. Dermatologists can employ automated analysis of these dermatoscopy images to aid diagnosis and decision-making. Deep learning techniques have addressed skin lesion classification (SLC) issues [9]. In contrast to conventional feature extraction methods, using feature selection for skin lesion identification and classification has become increasingly important in recent years. The most crucial step is segmenting the diseased area and choosing essential features from the raw data. Brightness, lesion borders, and color resemblances between the skin and the lesion in skin cancer images complicate and lower classification accuracy [10]. These factors make it necessary to classify skin cancer images in some processes, including image preprocessing techniques, extraction of important features, and classification. The research space identified in the mentioned studies can be summarized as follows: first, further research is needed to develop precise and efficient methods for detecting and classifying skin lesions. Various techniques have been proposed, such as automatic image processing, deep convolutional neural networks, and ensemble frameworks, but there is still room for improvement in accuracy, sensitivity, and specificity. To ensure their robustness and generalizability, method performance should also be assessed on bigger and more varied datasets. Second, methods for balancing and augmenting data have demonstrated the potential to enhance the classification of skin lesions. More thorough research is nevertheless required to evaluate these technologies' performance across a range of datasets and contexts. Furthermore, research into computationally efficient models is needed to develop reliable for skin cancer detection. Finally, a comparative analysis shows the superiority of ensemble models compared to traditional machine learning algorithms for skin lesion classification tasks, especially in improving performance on large and imbalanced datasets. This study fills this gap by proposing an approach that combines deep learning models and data augmentation for balance techniques to improve performance in skin lesion classification.

This work leverages the advantages of data augmentation to address the imbalance in skin types with a reduced set of images. By generating a well-balanced dataset, we aim to enhance the accuracy of skin classification. The primary contributions of this study are as follows:

- The medical skin lesion images are resized to reduce memory consumption and improve latency, improving latency, thereby enhancing the model's efficiency and applicability in real-world scenarios.
- We employed traditional data enhancement techniques were employed to overcome the imbalanced data between classes, which not only reduces overfitting but also improves the robustness of the model.
- An ensemble model based on IncepX-Ensemble was developed specifically to classify seven types of skin lesions based on a weighted average. As a result of this approach, the overall performance of classification is improved by leveraging the strengths of multiple models.
- In order to comprehensively assess the model's performance, we applied a variety of evaluation metrics and a confusion matrix. Through this comprehensive evaluation, a higher standard of performance validation is set in skin lesion classification.

II. RELATED WORK

Researchers have proposed numerous studies in recent years to detect early-stage skin cancer. Most of the latest research is included in this section. According to the World Health Organization (WHO) [11], 13.1 million people are expected to die from cancer worldwide by 2030, most of them in the United States, where skin cancer affects most people. As it penetrates quickly due to abnormal cell growth, skin cancer

often spreads to other parts of the human body and is a common human disease [12]. Various approaches have been tried in recent years. Presents and practices, with a focus on the health field classify skin lesions.

Skin cancer's high fatality rates have spurred global scientific efforts in its classification and detection. Theoretical literature on skin cancer is expanding, benefiting from ongoing advancements in science and technology. Evolving methods for skin cancer classification and diagnostics progress from single imaging to targeted classification, employing diverse methodologies like computer vision-based machine learning. These approaches integrate supervised learning and deep learning algorithms for reliable skin lesion classification. Notably, the use of Convolutional Neural Networks (CNN) has gained prominence, demonstrating advanced machine-learning applications in skin cancer classification. In the study [13], authors looked into how image pre-processing affected how well a saliency segmentation method worked for skin lesions. The results confirm that the CHC-Otsu approach performs as intended and achieves a 92% accuracy rate. Computer-aided diagnostic (CAD) [14] systems offer distinct advantages to radiologists today, with CAD being an integral part of contemporary medical imaging. These systems are great because of the advanced algorithms for fine-grained detection and classification. More recent studies on skin cancer detection, [15], [16] have demonstrated that using CAD systems can enhance the accuracy of diagnosis and shorten analysis time. It is widely used to generate synthetic samples and balance datasets using methods such as Synthetic Minority Oversampling Technique (SMOTE) and Adaptive Synthetic Sampling Approach(ADASYN). Traditional image-based data augmentation includes flipping, rotating, etc. as well as generative models for synthetic medical data generation.

A. CNN METHODS FOR THE CLASSIFICATION OF SKIN LESIONS

In recent years, machine learning technology based on DCNN has been widely established as a cutting-edge technology for medical video classification. Some examples of such technologies are GoogleNet [17], Microsoft ResNet [18], DenseNet [19], MobileNet [20] and EfficientNet [21]. Model selection is an area of notable growth. These methods eliminate the need for a specific base model by prioritizing the selection of unsupervised model types for detection. The accuracy of the classified model increased after it was trained on a specific dataset. The authors [22] proposed RDDNN (Residual Deep Convolutional Neural Network) for skin lesion diagnosis and trained and tested six well-known skin cancer datasets. The experimental procedure begins in two steps. One is to use the original image without preprocessing or segmentation, the other is to use the segmented image to evaluate network performance under different conditions. Hosny et al. [23] proposed a deep inherent learning technique to classify seven types of skin lesions. Several explanatory strategies were used to validate the proposed detailed unique learning. A residual deep convolutional neural network was suggested by the author [24]. Their method primarily modifies the multiple convolution filters for cross-channel correlation and multi-layer feature extraction by shifting the dot product filter rather than the filter along the horizontal axis. In addition to using the ISIC-2019 and ISIC-2020 datasets, the suggested solution solves the issue of imbalanced datasets by converting the picture and label dataset into a vector of images and weights.

In the study by Polat and Koc [25], raw skin lesion images were used as input to classify multi-class skin diseases. They claimed that combining a CNN and a one-versus-all approach obtained auspicious results in skin disease classification from dermatoscopy images. Lima et al. [26] proposed combining CNN methods with the Squeeze-and-Excite mechanism in a residual block. The model is trained with the data from scratch for the different classification problems. The findings demonstrate that it is significantly more likely than other ways to have a free and favourable influence. A hybrid-CNN classifier for recognizing numerous skin diseases was proposed by Hasan et al. [27] To produce better depth feature maps of the lesion, Hybrid-CNN fuses three different feature extractor modules. Tang et al. [28] suggested a multi-stage, multi-modal learning system to classify skin lesions with many labels. Two phases make up the suggested model. In the first phase, they construct Level 1 of the fusion, which unifies the feature-level representation of images. They propose a different fusion, training a Support Vector Machine (SVM) cluster with patient meta-data information and multi-label prediction data from the first stage. The patient's meta-data would then be added in the second stage. The proposed network classifies skin lesions with 78.5% accuracy using multiple labels. A two-module structure with lesion segmentation and classification was suggested by Khan et al. [29]. Binary images from the 16-layer CNN model and the improved HDCT structure were trained for lesion segmentation.

B. DATA AUGMENTATION TECHNIQUES

The problem of limited labeled data and class imbalance are due to color space transformations (saturation, contrast, brightness, tone adjustment, etc., of training images), geometric transformations (mirroring, cropping, rotation, translation, etc.), blending images, random deletion, etc. This can effectively increase the size of the training data set [30]. Perez et al. [31] provided a data augmentation scenario for skin lesion classification trained with a CNN classifier. Their study showed that geometric and color space variations can improve classification performance. However, shuffling and randomly deleting images could not able to produce better results. This is because these two data augmentation methods cannot ensure that the transformed image retains the relevant skin lesion features of the original image. Skin

lesion classification performance can be improved through data augmentation techniques such as geometric and color space modifications. Since the distribution of the transformed images is similar to the distribution of the original images and does not add any new visual elements to the training set, there is little performance improvement [32], [33]. Training deep learning models is challenging due to the need for labeled and class-imbalanced datasets. Transfer learning was employed by Qin et al. [32], to enhance classification performance using data augmentation technology based on GANs. To improve the categorization accuracy of cells acquired from bone marrow aspirate smears, Hazra et al. [34] advocated using GAN-based data for the classification purpose.

C. ISIC DATASET DISCUSSION

Jasil and Ulagamuthalvi [35] used inceptionv3, VGG16, and VGG19 to classify skin lesions, including the transfer learning idea. On the ISIC dataset, they attained accuracy rates of 74% with InceptionV3, 77% with VGG16, and 76% with VGG19. Alhudhaif et al. [36] employed the ISIC dataset to evaluate six transfer learning networks for the classification of skin cancer. Augmentation was applied to images of smaller classes to address imbalances in class sizes within the ISIC dataset. The outcomes revealed that augmentation is impactful for achieving classification success, showing high accuracy and F1 scores with fewer false negatives. Among the transfer learning networks investigated, the Modified DenseNet121 emerged as the most successful model, achieving an accuracy rate of 98.35%. Chatterjee et al. [37] proposed an integrated, computer-aided method that has been employed to identify each of these skin diseases using RFE-based layered structured multiclass image classification. Hosny et al. [38] proposed a method for classifying skin lesions that combined transfer learning with AlexNet that had already been trained. We randomly initialize the weights of the final three replacement layers using the parameters of the original model as beginning values. The most recent publicly available dataset, ISIC 2018, was used to test the suggested approach, which has a 98.7% accuracy rate. An MCML classification technique was proposed by Hameed et al. [39] for a multi-class skin disease categorization. Both sophisticated deep learning techniques and conventional machine learning were employed in the MCML classification algorithm. Using a small dataset, the algorithm demonstrated 96.4% accuracy in testing. They also claimed that the proposed method could significantly enhance classification performances in multi-class skin datasets. A dual attention-based network for skin lesion categorization with auxiliary learning was proposed by Wei et al. [40] The spatial and channel attention modules are a part of the dual attention system. When trained on the ISIC 2018 and ISIC 2019 datasets, the suggested model has a precision of above 80%. Transfer learning is the process of using a model that has already been trained on huge data, and now we are trying to teach a new model with some new features rather than starting from scratch. Chatterjee et al. [41] proposed an Alzheimer's disease classification mechanism using a voting ensemble approach as a smart way to investigate the type of disease. A Convolutional Neural Network (CNN) model for the classification of skin diseases was proposed by Alwakid et al. [42]. They classified skin illnesses with a 92% accuracy rate using a collection of 10015 images taken from the ISIC 2019 dataset.

D. DATA IMBALANCE CHALLENGES

The problem of imbalanced data has been tackled in a number of ways, especially in the area of object detection when the background (majority) and foreground (minority) classes are not equally represented. In the paper by authors [43], proposed accurate automated melanoma classification using dermoscopic images. To solve the problem of interclass similarity, the authors of this article developed a unique, lightweight DeepCNN-based framework to distil information in the context of melanoma classification. In the study, they proposed to apply cost-sensitive learning with loss of focus to solve the critical class imbalance problem and achieve higher sensitivity values.

To ensure accurate predictions, Akilandasowmya et al. [44] using the ResNet50 method (SCSO-ResNet50), used a novel approach called sand cat swarm optimization to distinguish between known features and deeply hidden features. Apply enhanced harmony search (EHS) techniques to minimize data dimensions and maximize properties. Ensemble classifiers can help detect cancer early. This research proposes a method to enhance the DCNN classification performance on imbalanced datasets for multi-class medical images. The authors [45] suggest utilizing optimal domination of weights across asymmetric classes with majority and minority categories to solve the issue of imbalanced samples, as well as a robust feature extraction strategy for improved training and loss function optimization.

E. ENSEMBLE LEARNING BASED

In recent years, the complexity and drawbacks of real-world problems have increased, requiring the development of new advanced algorithms, transfer learning techniques, meta-leaning models (ensembles), and more reliable models. The authors [46] used a meta-learning algorithm with a stacking ensemble method to achieve better performance. Using a weighted average ensemble learning model that integrated multiple transfer learning models, Rahman et al. [47] identified 18,730 lesion pictures from the ISIC 2019 and HAM10000 datasets. The ensemble model with a weighted mean achieved 94%, whereas the ensemble model with a basic mean achieved 93%. One of the models proposed by the study authors is a multiscale deep ensemble [48]. To extract more features, the proposed model precisely cuts out the lesion area. They then linked the cropped image's features to several metrics to classify the results. As a result, HAM10000

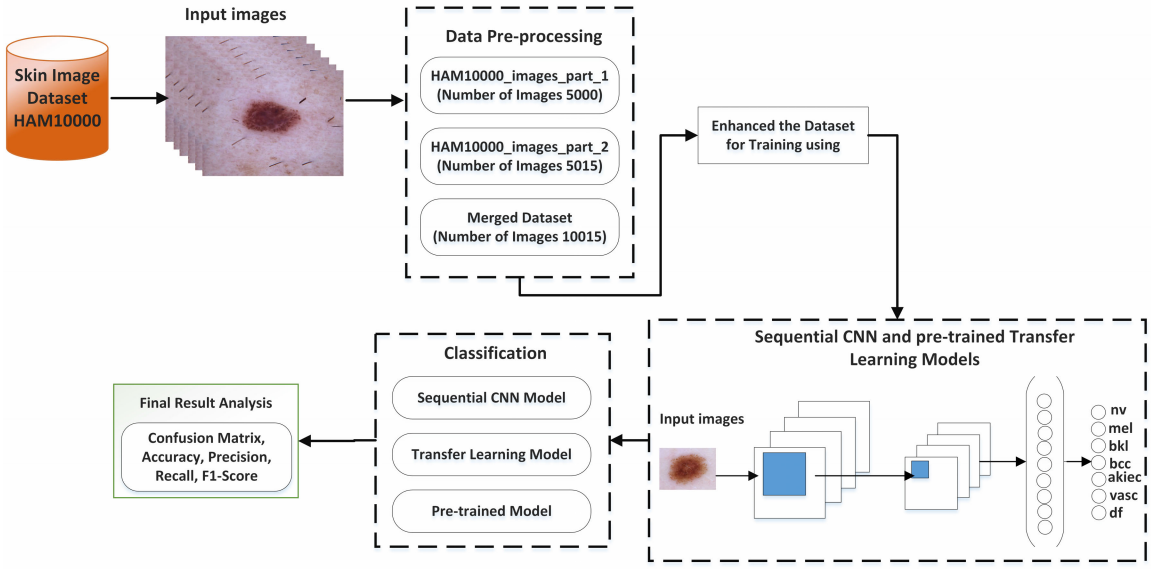


FIGURE 1. Flowchart of the proposed multi-class skin lesion classification.

fragments of the training dataset are reported to outperform other approaches.

III. METHODOLOGY

This work presents a multiclass skin cancer classification methodology using a pre-trained transfer learning model with best-selected features. Fig. 1 provides a complete flowchart of the methodology. Additionally, the study's data collection and the image preprocessing techniques applied to it. The proposed IncepX-Ensemble model utilized for the classification problem is briefly presented, and data augmentation was done to balance the data set. We integrated the inceptionV3 model with the Xception model, which effectively made the ensemble model improve performance and computation efficiency. In this research, we introduce an advanced model that leverages weighted average ensemble learning to classify seven different types of skin lesions. The model combines the strengths of two well-known deep neural networks, InceptionV3 and Xception, as its base components. InceptionV3 is known for its efficiency in handling complex visual tasks through its unique architectural features. Xception, on the other hand, employs depthwise separable convolutions to enhance performance and accuracy. By integrating these two networks, our ensemble model named the IncepX-Ensemble model aims to improve classification accuracy. This approach allows us to effectively utilize the diverse capabilities of both networks to achieve superior results in skin lesion classification. The dataset was improved using a data enhancement technique after the preprocessing stage. The proposed model was applied to both the unbalanced original dataset and the datasets that had been balanced using balancing procedures. Following the training, experiments for testing and validation were conducted along with the classification procedure.

A. DATA ANALYSIS

Dermatoscopy can improve the doctor's view of the skin, helping to diagnose different skin conditions and consider suspicious skin lesions. Compared to ocular inspection, this approach is frequently utilized to improve the identification of benign and malignant pigmented skin lesions [49]. One of the most extensive datasets of 10015 dermoscopic images gathered over 20 years is the HAM10000 (Human Against Machine) dataset. The information was collected from research labs at several universities, the dermatology department at the Medical University of Vienna in Austria, and Cliff Rosendahl's skin cancer clinic in Queensland, Australia [50]. All records of the HAM10000 data set are archived in the Harvard Dataverse. A total of 10,015 skin lesion images from seven main kinds of skin lesions make up the HAM10000 dataset. Fig. 2 displays the sample images for each skin lesion type. Fig. 3 exhibits the distribution of samples across different classes in the original dataset using a bar representation. The distribution of classes on the data set is disrupted by the imbalanced number of samples belonging to the Melanocytic nevi (nv) class, as seen in Table. 1. A high-class imbalance exists in the dataset, with more than two-thirds of the images falling under the nv class.

Table 1 provides a detailed breakdown of the distribution of image samples across the HAM10000 dataset for the most crucial diagnostic classes. The data set includes 6705 images of melanocyte nevi (nv), 1113 images of melanoma (mel), 1099 images of benign keratosis-like lesions (bkl), 514 images of basal cell carcinoma (bcc), 327 images of actinic keratosis (akiec), and 142 images of Vascular lesions (vasc) images and 115 dermatofibroma (df) images. However, to effectively train a deep learning-based model, we need sufficient balanced data. Intentionally

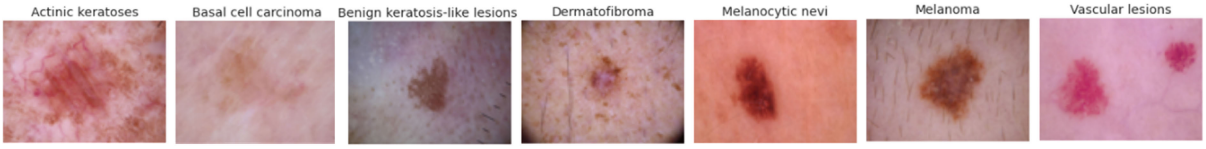


FIGURE 2. Sample of each skin type are shown in the figure.

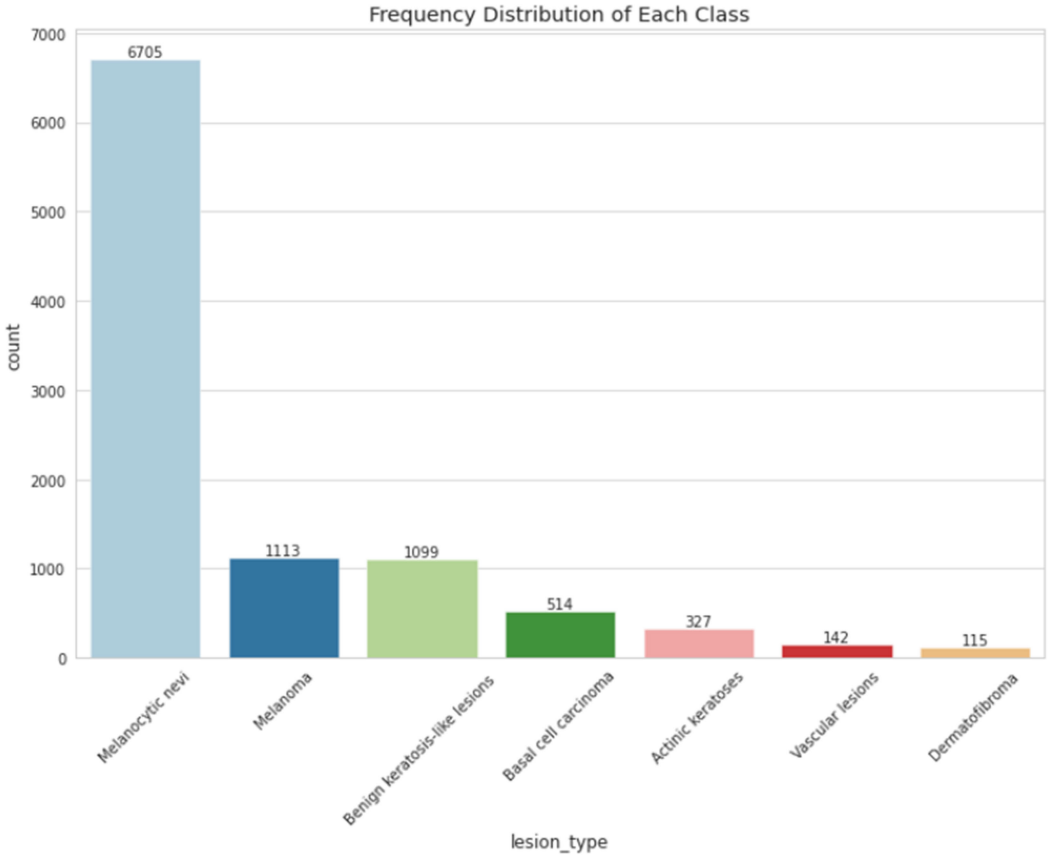


FIGURE 3. Number of instances distribution across the seven classes is shown in the figure.

TABLE 1. HAM10000 dataset distribution of the images among the seven main skin lesions classes.

Class Name	Abbreviate Diagnostic Category	No. of Images	Percentage
Melanocytic nevi	nv	6705	66.95%
Melanoma	mel	1113	11.11%
Benign keratosis-like lesions	bkl	1099	10.97%
Basal cell carcinoma	bcc	514	5.13%
Actinic keratosis	akiec	327	3.27%
Vascular lesions	vasc	142	1.42%
Dermatofibroma	df	115	1.15%

generating the necessary data helps achieve data balancing, preventing biased model training. This precaution is taken when the imbalanced data could lead to ongoing bias in model training, particularly favouring classes with a substantial number of samples. In the pie chart diagrams, the distribution of diagnosis categories in each class is depicted in the left side pie chart, which is shown in Fig. 4. The right side of Fig. 4 shows the distribution of the diagnosis method. More than 50% of the lesions in the images were confirmed histopathologically, and the remaining cases were confirmed by follow-up examinations, expert consensus, or in vivo confocal microscopy. Each lesion's diagnosis was verified using a particular method. A total of 5,340 (53.3%) lesions were confirmed histopathologically, 3,704 (37.0%) by

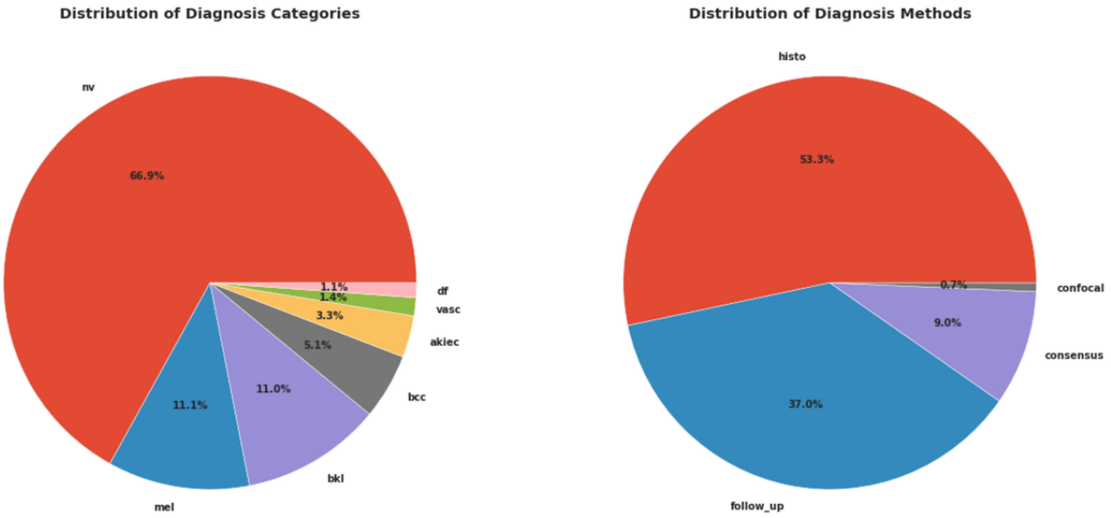


FIGURE 4. Pie charts representing the distribution of skin lesion data. As illustrated in the left graph, different diagnosis categories (e.g., nv, bkl, df, vasc) are distributed by percentage, while different diagnosis methods (e.g., histo, follow-up) are shown in the right graph.

follow-up, 902 (9.0%) by expert consensus, and 69 (0.7%) were confirmed based on ground truth.

1) IMAGE PREPROCESSING

Each image in the dataset has a dimension of 600 × 450 pixels. We expanded the dataset by using rotation and flipping as part of conventional image-enhancing methods. We artificially increased the size by rotating the dataset and flipping it horizontally and vertically. Because neural networks need a lot of labelled data to train, dataset size has typically been a problem in the medical field. To overcome the data imbalanced issue, we balanced the dataset by using data augmentation to train the model with balance. Data augmentation is helpful when a significant data imbalance is distributed among the dataset's classes. Data augmentation may impact model training if the data is unbalanced or very small. This can lessen the deep neural network's overfitting. To address the problem of data imbalance in most of the classes in the HAM10000 dataset, we applied data augmentation techniques.

The precise kind of data augmentation method employed in the investigation is described in depth. A variety of data augmentation methods, such as rotation and inversion, were used in our methodology. I specifically reversed the image both horizontally and vertically and rotated it 90 degrees clockwise, 90 degrees counterclockwise, and 180 degrees. These techniques help increase the size of the training data set, preventing overfitting and exposing the model to different transformations of the same image, thereby increasing the robustness of the model. Our methodology included several key steps. First, the images are read from the dataset. I then created a new image by rotating and flipping using the ''producer_new_img'' function. For images from classes 1 to 6, additional augmented images were created to balance the class distribution. Next, we extracted the labels corresponding to the images and assigned them to the augmented images as well. To illustrate the impact of the enhancement technique, Fig. 5 shows the frequency distribution of classes before and after enhancement.

B. PROPOSED MODEL

A powerful technique to accelerate model development is transfer learning in the field of deep learning. Compared to creating a model from scratch, transfer learning drastically cuts down on computational resources and training time. Moreover, it can be very helpful when dealing with small datasets. To train the model, the image classification field often needs a sizable collection of tagged images. But by using the advantage of transfer learning, we may ''fine-tune'' a pre-trained model trained on ImgaeNet. Compared to building a model from scratch, this method uses less time, which makes it a useful tool for applications requiring small datasets. Transfer learning techniques are utilized to enhance the performance of machine learning algorithms using labeled data. These efforts involve learning and applying one or more source tasks to improve learning in related fields. Transfer learning has been studied as a machine learning process to address various problems. It includes pre-training models on large datasets and subsequently retraining them at different levels on smaller training sets. Adjustments to the initial layer of the pre-trained network can be made if necessary. Fine-tuning the parameters of the model's final layer allows it to learn from new datasets. According to the new task, pre-trained models are retrained with a smaller new dataset, modifying the model weights accordingly. This means that newly developed neural network parameters are not built from scratch. Although deep learning algorithms can achieve higher functionality or performance for many

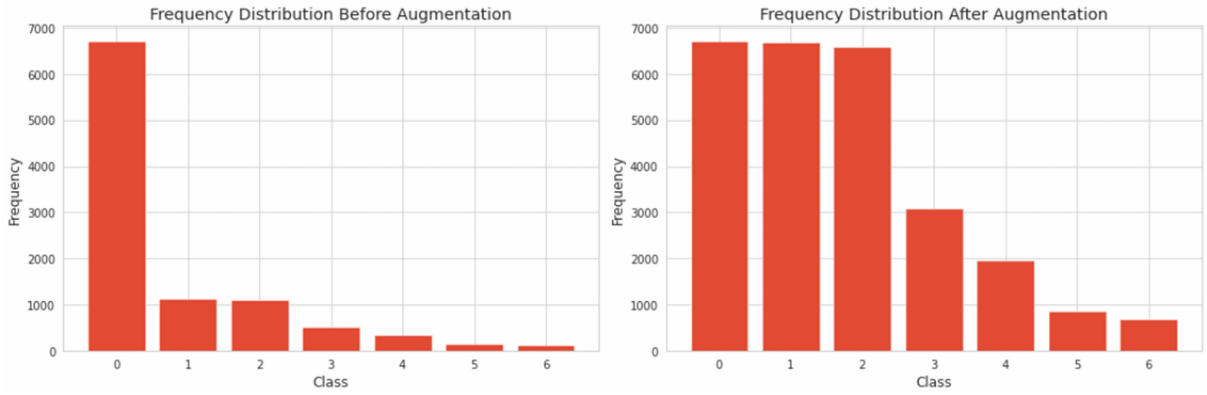


FIGURE 5. The figure depicts the frequency distribution of classes before and after data augmentation. The diagram on the left shows the frequency distribution of the original class labels before data expansion. The graph on the right shows the frequency distribution after applying data augmentation techniques.

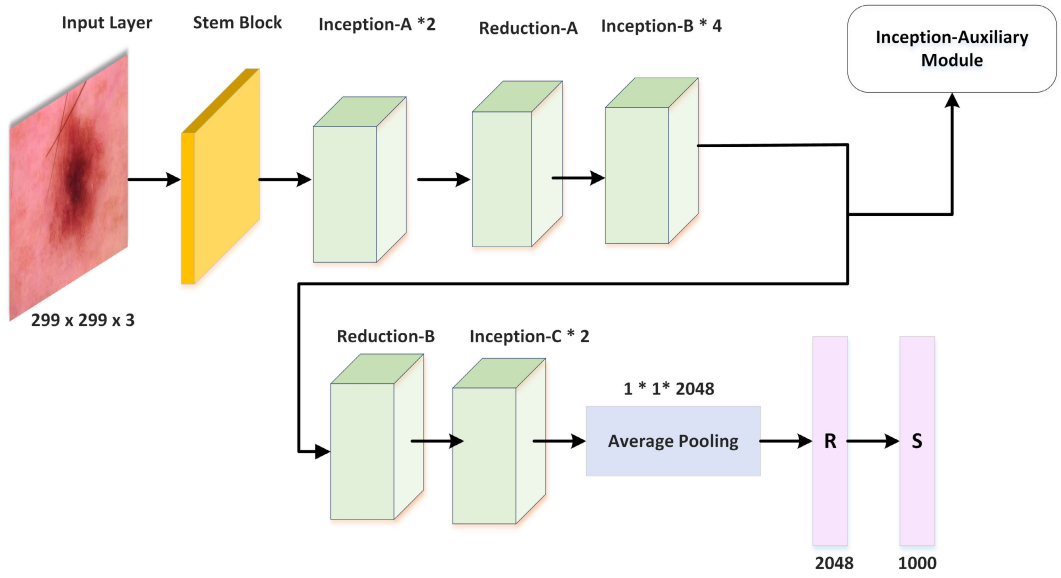


FIGURE 6. Network architecture of InceptionV3.

problems, they require extensive data and training time. Thus, reusing pre-trained models for similar tasks can be highly beneficial. In our approach, we used two pre-trained models: InceptionV3 and Xception. These models, initially pre-trained on the ImageNet [52] dataset, were fine-tuned using the skin lesion dataset.

The ImageNet dataset, comprising millions of images, was used to train the Inception [53] and Xception [54] network models. The InceptionV3 network, with 48 layers, and the Xception network, with 71 layers, both require input images of size 299 × 299 × 3 pixels. Fig. 6 and Fig. 7 illustrate the structural layouts of InceptionV3 and Xception, respectively.

We developed a special image resizing method that takes into account the different sizes of the input images while preserving important features. The image size was reduced to 299 × 299 pixels to retain important visual information while maintaining computational efficiency. After much testing, we chose this dimension so that resized images retain a high level of information necessary for accurate classification.

We also used data augmentation methods such as flipping and rotating to add variety and creativity to our training set. By mimicking the variance found in the real world and avoiding overfitting, this extension helps improve the generalization of the model. We use these changes to improve the model's detection and classification efficiency for new instances by ensuring that the resized images fit the model's input size while preserving their original properties and variants.

InceptionV3 addresses common issues such as congestion and yield by employing asymmetric filters and bottlenecks, and replacing larger filters with smaller ones, leading to effective outcomes. This architecture optimizes the network's

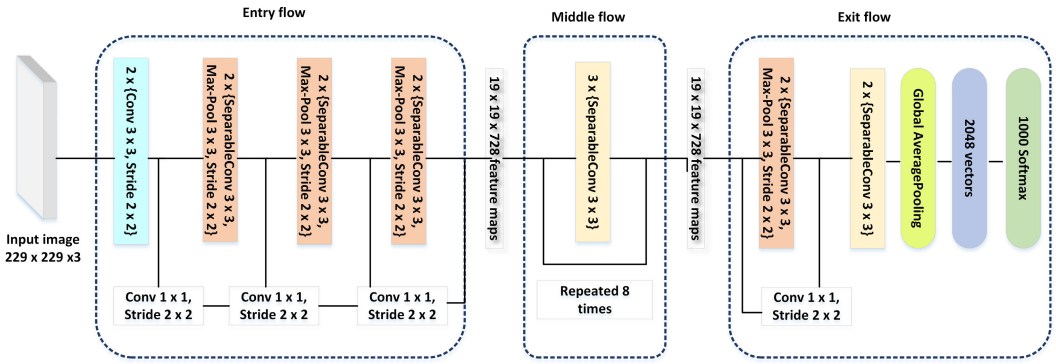


FIGURE 7. Network architecture of Xception [51].

performance by managing computational complexity and enhancing feature extraction efficiency. The design includes factorized convolutions and aggressive regularization, which contribute to its robustness in handling complex visual tasks. On the other hand, Xception is designed to be more efficient and less complex. It achieves more precise and efficient results by separately applying cross-channel and spatial correlations, thus improving the model's accuracy and speed. Additionally, the Xception model utilizes depth-wise separable convolution and employs cardinality to develop more effective abstractions, enhancing its performance. This separation allows for the creation of more nuanced representations, making Xception particularly effective for various image classification tasks. By leveraging these advanced architectures, both InceptionV3 and Xception achieve high accuracy on large-scale image classification benchmarks, demonstrating their suitability for diverse applications in computer vision.

InceptionV3 and Xception have been successfully applied in various research areas, including object detection, image segmentation, and medical image analysis. Their high performance on the ImageNet dataset has led to widespread adoption in tasks requiring high accuracy and robustness. InceptionV3's factorized convolutions and aggressive regularization make it well-suited for handling complex visual tasks while managing computational complexity. On the other hand, due to using depth-wise separable convolutions both efficiency and performance are improved by Xception - leading to accomplished results for various standard image classification tasks. Both of these models benefit from being pre-trained on large amounts of data on ImageNet which makes them a great option for transfer learning in low-data domains. These models can be computationally challenging, though, which makes it demanding to train or draw conclusions from them. Their architectures are intricate, and it can be challenging to reuse or fine-tune a pre-trained model without considering the unique limitations of your intended application. We decided on these models because their modern architectures and state-of-the-art performance on large-scale image classification tasks prepare them well for our task of increasing the accuracy of skin lesion classification. Combining the strengths of the proposed IncepX-Ensemble modes by sharing the free features of Xception and InceptionV3 and maximizing their benefits.

First, we inserted pre-trained ImageNet weights into the InceptionV3 and Xception models, leaving out the top layer. After that, we concatenated each model's output and added a global spatial average pooling layer to create a composite feature vector. To create the final prediction, a softmax layer with seven classes was added after a dense layer with 1024 neurons and ReLU activations. To increase its capacity for generalization, this ensemble model was assembled using the Adam optimizer and trained utilizing data augmentation methods.

1) ENSEMBLE LEARNING

Using ensemble learning, multiple models can be combined to improve performance and computation efficiency. Ensembles of deep neural networks always perform better than single models, regardless of the number of models. Each model was weighted equally in this study, using average ensemble learning.

$$P = \frac{\sum A_i}{B} \tag{1}$$

The probability of model i can be calculated from equation 1, and the total number of models is B.

Deep learning models often yield different results due to their varying architectures and levels of complexity. By weighting supervised learning models based on their performance, we can ensure that each model's output is optimized to the fullest extent possible. However, identifying the optimal combination of model weights is a challenging task. To address this issue, we utilized the grid search technique, as shown in Fig. 8. We explored a total of 2000 different weight combinations. The search process continued until all possible combinations were evaluated. Ultimately, this method gave us the optimal weights, maximizing our evaluation metrics.

TABLE 2. Model hyperparameters summary.

Activation	Optimizer	Batch Size	Epochs	Learning rate	Loss function
ReLU	Adam	32	100	0.0001	categorical_crossentropy

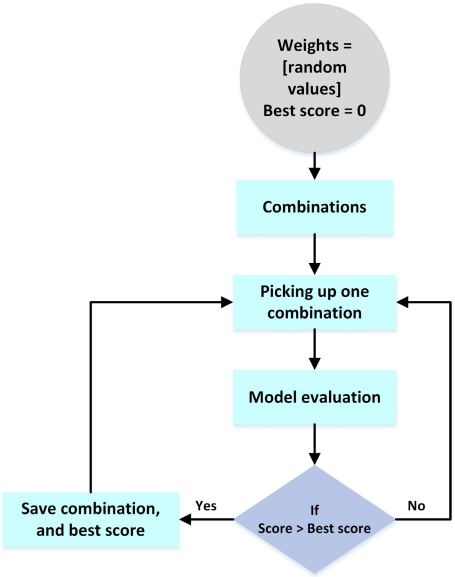


FIGURE 8. Weights are determined by grid search.

In unsupervised learning environments where the optimal outcome is unknown, clear tactics must be implemented. Weights can be assigned according to the consistency and agreement between models using methods such as mutual information and consensus clustering. In addition, the self-learning technique of the pseudo-labeler serves as an alternative to performance evaluation and supports weighting procedures in the event of missing labeled data.

Taking advantage of ensemble learning technique to improve the performance and robustness of machine learning models, is the main contribution of this study. Combining several models helps us take advantage of their strengths and minimize their weaknesses. This approach often provides better performance than a single model can achieve. Specifically, in this study, we used an average ensemble method in which each model contributes equally to the final prediction. Techniques for grid search are very useful since they make it possible to explore the hyperparameter space systematically. Finding the ideal weighting combination is computationally demanding, but the results are well worth the effort. Gives the best-performing models more weight, enabling the ensemble model to utilize each model to its fullest potential. The grid search method for determining the ideal model weights is depicted in Fig. 8. We rigorously assess every conceivable combination, starting with random values, to identify the optimal combination. In conclusion, grid search methods in conjunction with ensemble learning strategies offer a potent means of maximizing model performance. By doing so, extremely effective ensemble models are produced, maximizing the benefits of each model.

TABLE 3. Comparison of the different model performances with the original imbalanced dataset.

Model	Accuracy	Precision	Recall	F1-score
CNN	0.82	0.81	0.81	0.83
Resnet50	0.83	0.83	0.82	0.83
InceptionV3	0.85	0.85	0.83	0.83
proposed model	0.86	0.86	0.87	0.86

IV. EXPERIMENTAL RESULTS ANALYSIS

The detailed outcomes of the experiments are discussed, and a performance analysis of the results is presented in this section. A description of the evaluation metrics and details of system implementation is provided. In this section, the experimental outcomes are deployed with the help of tables, confusion matrices, and accuracy-loss plots. All machine learning models were trained and tested on a Ubuntu system, with GEFORCE RTX 4080 GPU, and 32GB of RAM. The environment has Python version 3.7.7.

A. EVALUATION METRICS FOR PERFORMANCE ANALYSIS

The suggested model is trained over 100 epochs using a batch size of 32 and Adam as the optimizer. The model is trained using an initial learning rate 0.0001 while efficiently using the number and kernel sizes. We chose this learning rate based on preliminary experiments and existing literature suggesting that a lower learning rate can improve training stability. Nevertheless, we acknowledge the risk of training slowing down if this low learning rate is used initially. Table 2 summarises the model hyperparameters in general. In the model training, the activation function ReLU is utilized. We split the total dataset 80% as training data and 20% as testing data.

Calculated metrics such as accuracy, precision, recall, and f1-score are used to assess how well the trained model performed with test samples. Images of skin lesions without enhancement serve as testing samples. Knowing the model's performance for hypothetical samples is the goal here. The metrics stated in Equations 2–5 were used to assess how well each system performed when classifying images of skin lesions for the HAM10000 dataset.

$$\text{Accuracy}(acc) = \frac{T_P}{T_P + T_N + F_P + F_N} \tag{2}$$

$$\text{Precision}(pre) = \frac{T_P}{T_P + F_P} \tag{3}$$

$$\text{Recall}(rec) = \frac{T_P}{T_P + F_N} \tag{4}$$

$$F1 - \text{score} = \frac{2 * T_P}{2 * T_P + F_P + F_N} \tag{5}$$

These formulas include true positive T_P and true negative T_N, representing various correctly detected positive and negative samples. Additionally, the terms false positive F_P

TABLE 4. Comparison of the different model performances with data augmentation.

Model	Accuracy	Precision	Recall	F1-score
Data augmentation with CNN	0.919	0.919	0.921	0.922
Data augmentation with Resnet50	0.935	0.935	0.933	0.935
Data augmentation with InceptionV3	0.957	0.958	0.953	0.959
Data augmentation with proposed model	0.988	0.988	0.989	0.989

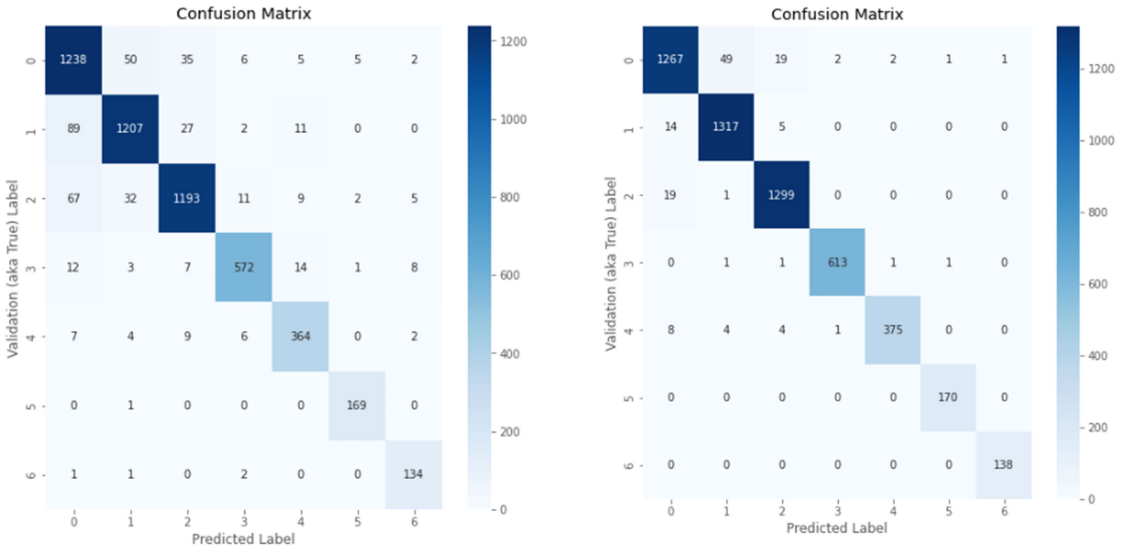


FIGURE 9. Confusion matrices showing the results of skin lesion classification. Left: Results from the CNN model predictions. Right: Results from the proposed IncepX-Ensemble model predictions.

and false negative F_N describe the quantity of incorrectly detected positive and negative samples.

Four comparisons were made to demonstrate the proposed model's effectiveness. First, we evaluated the classification performance using the CNN models with the IncepX-Ensemble model when data augmentation was used. The significance of the proposed model was thoroughly assessed through a comprehensive evaluation that included comparisons with and without data augmentation, as highlighted in Tables 3 and 4. It is important to emphasize that the results obtained indicate a notable superiority of the proposed model over alternative counterparts in terms of accuracy, specificity, and sensitivity. This heightened performance underscores the robustness and effectiveness of our model in addressing the challenges posed by the dataset. Data augmentation further enhances the model's ability to generalize and perform well across diverse scenarios, contributing to its superiority over alternative approaches.

Computation:

- **Sequential CNN Model:** Sequential CNN model: This model includes standard convolution and pooling layers, resulting in a relatively simple calculation process. The computational complexity of a single CNN is generally $O(n \times m \times k \times d)$, where n is the number of filters, m is the filter size, k is the number of input channels, and d is the network depth.
- **IncepX-Ensemble Model:** Ensemble models combine multiple base models to increase the computational load through additional forward and backward propagation during training. The combined complexity can be approximated by $O(n \times m \times k \times d \times L)$. Here L is the number of models in the set. However, parallel processing functions reduce the overall computational effort and enable efficient training.

B. COMPARISON WITH CONFUSION MATRIX

A confusion matrix is a table commonly used to describe the performance of classification models. Compare the actual target value with the model predicted value.

- **Number of diagonals:** The confusion matrix's diagonal elements show the instances in which the actual and expected labels coincide. The amount of accurate predictions for each class is represented by this value. The number at position (i, i), for instance, shows how many times the model accurately predicted class i. When the diagonal value is high, it indicates that the model is successful in accurately classifying instances of a given class.
- **Numbers on the Off−Diagonal:** Off-diagonal elements represent cases where the predicted label does not match the actual label. This value indicates the number of incorrect predictions. For example, the number at the (i,j) position (where $i \neq j$) indicates how many times

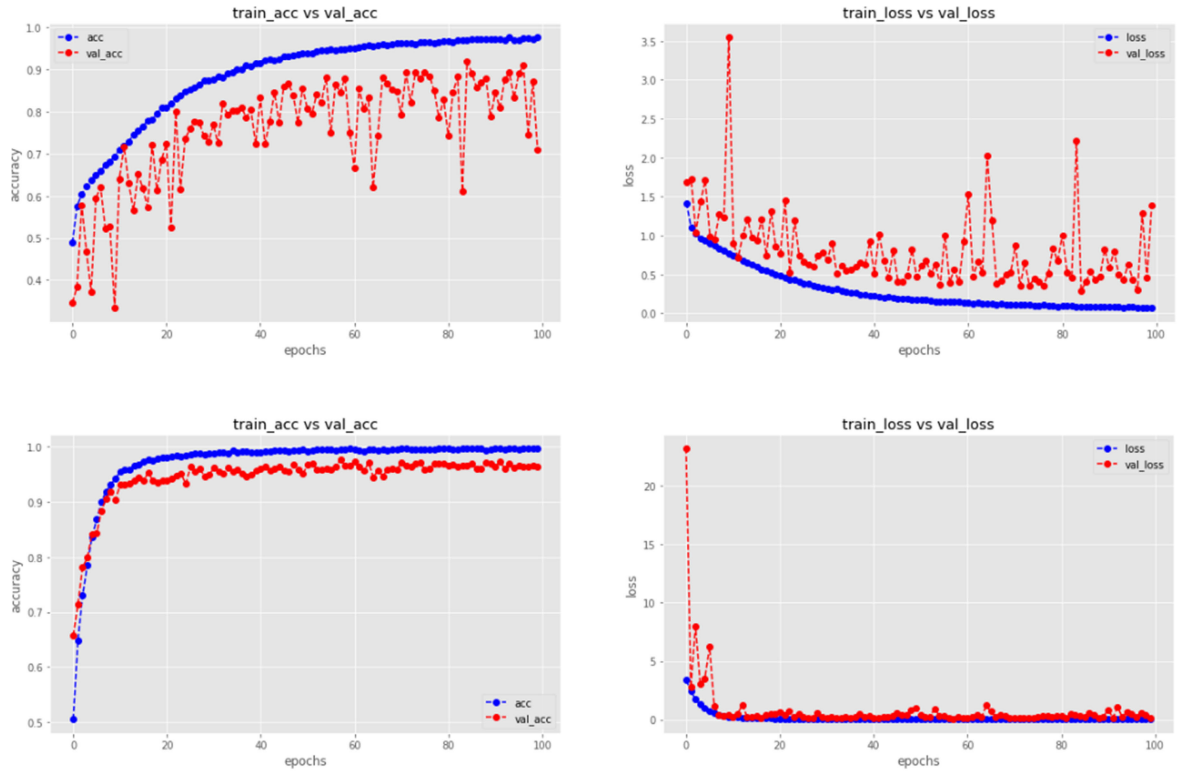


FIGURE 10. Training and validation loss and accuracy results over time for both models. The top row shows the learning curves for the CNN model, while the bottom row shows the learning curves for the IncepX-Ensemble model.

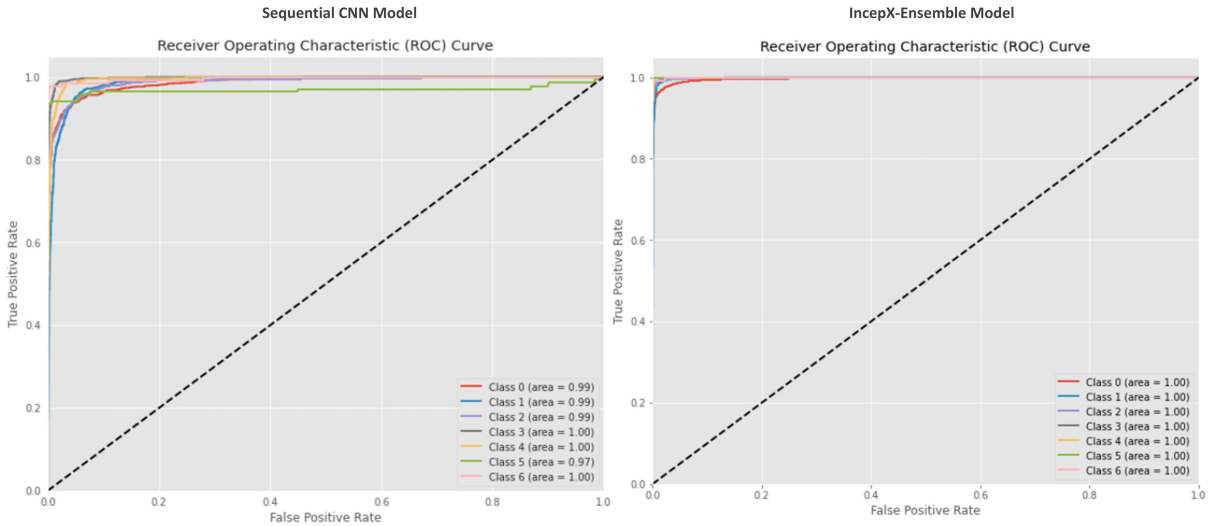


FIGURE 11. Receiver Operating Characteristic (ROC) curves comparing the performance of the Sequential CNN model (left) and the IncepX-Ensemble model (right) on the skin lesion classification task.

the model incorrectly predicted class j even though the actual class was i. A lower value on the off-diagonal line is preferable, as it indicates less misclassification.

We can learn more about the errors that occur in our model by looking at the confusion matrix. For instance, a high value for a specific off-diagonal line suggests that the model frequently mixes together different classes. This data aids in identifying particular model flaws and serves as a roadmap for future enhancements.

A confusion matrix is invaluable in evaluating the performance of a multi-class image classification model, providing a detailed breakdown of the model's success in classifying instances for each category. Fig. 9 shows the class-wise predictions using the confusion matrices. On the left is the

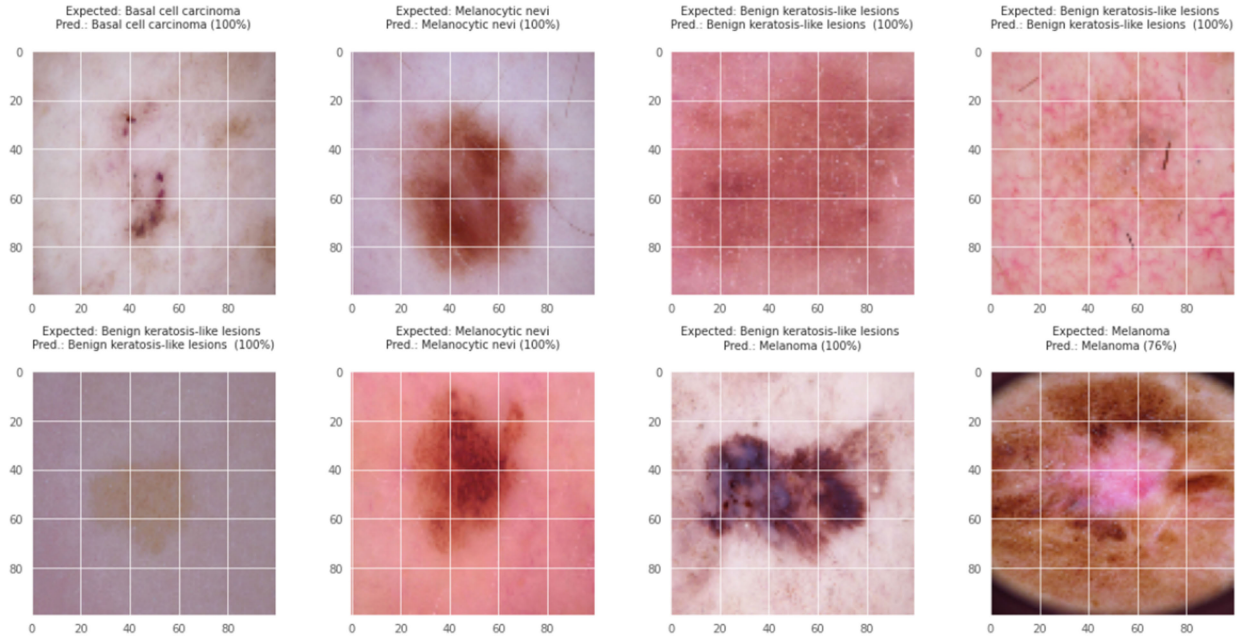


FIGURE 12. Samples of misclassification in skin lesion classification. The image depicts instances where the model's predictions diverge from the actual skin lesion types.

TABLE 5. Comparison of proposed model performance using HAM1000 dataset with some existing models.

Model	Dataset	Accuracy
Darknet-53 [55]	HAM10000	95.8%
EfficientNet [56]	HAM10000	84.3%
MobileNet + LSTM [57]	HAM10000	85.3%
DenseNet201 network [58]	HAM10000	92.8%
DenseNet169 [59]	HAM10000	91.1%
VGGNET-16 [60]	HAM10000	85.6%
Proposed Model	HAM10000	98.8%

predicted outcome based on the CNN model, and on the right is the predicted outcome based on IncepX-Ensemble.

In the confusion matrices, label 0 represents melanocytic nevi (nv), label 1 represents Melanoma (mel), label 2 represents Benign keratosis-like lesions (bkl), label 3 represents Basal cell carcinoma (bcc), label 4 represents Actinic keratoses (akiec), label 5 represents Vascular lesions (vasc) and label 6 represents Dermatofibroma (df). The predicted classification result with the CNN model, the number of misclassified images for each class were, class 0 with 103 misclassifications, class 1 with 129, class 2 with 126, class 3 with 45, class 4 with 28, class 5 with 1, class 6 with 4. The proposed IncepX-Ensemble model improved these results: class 0 had 74 misclassifications, class 1 had 19, class 2 had 20, class 3 had 4, class 4 had 17, and classes 5 and 6 achieved 100% accuracy.

Table 5 presents a comparative analysis of the proposed model's performance accuracy with several existing models using the HAM10000 dataset. The table outlines each model, including Darknet-53, EfficientNet, MobileNet + LSTM, DenseNet201 network, DenseNet169, and VGGNET-16,

TABLE 6. Training time comparison.

Model	Training Time (seconds)
Sequential CNN	12,900
IncepX-Ensemble	12,100

along with their corresponding accuracies. Notably, the proposed model achieves an impressive accuracy of 98.8% on the HAM10000 dataset, surpassing the performance of all other models listed. This considerable accuracy underscores the efficacy of the proposed model in accurately classifying skin lesions. Furthermore, it is worth highlighting that the proposed model achieves this high level of accuracy while maintaining a minimal number of trainable parameters and requiring a short calculation time. Such efficiency and effectiveness make the suggested model a promising candidate for skin lesion classification tasks, demonstrating its potential to contribute to medical image analysis and diagnosis advancements.

We have recorded over 100 generations of training time spent on both the Sequential CNN model and the proposed IncepX ensemble model. The results are summarized in Table 6. Training the sequential CNN model took 12,900 seconds and training the IncepX ensemble model took 12,100 seconds. We found that the IncepX ensemble model had slightly faster training times, despite the added complexity of combining multiple models. Thanks to the efficient parallelism capabilities of ensemble approaches, multiple models can be trained simultaneously to optimize overall calculation time.

Fig. 10 displays training and validation data's accuracy and loss curves. It shows the training history of the CNN and the

TABLE 7. A class-wise classification report comparing the sequential CNN model and IncepX-Ensemble.

	Sequential CNN model				IncepX-Ensemble Model			
Class Name	**Precision**	**Recall**	**F1-score**	**Support**	**Precision**	**Recall**	**F1-score**	**Support**
Melanocytic nevi-0	0.89	0.92	0.90	1341	0.97	0.94	0.96	1341
Melanoma-1	0.93	0.90	0.92	1336	0.96	0.99	0.97	1336
Benign keratosis-like lesions-2	0.94	0.90	0.92	1319	0.98	0.98	0.98	1319
Basal cell carcinoma-3	0.95	0.93	0.94	617	1.00	0.99	0.99	617
Actinic keratoses-4	0.90	0.93	0.92	392	0.99	0.96	0.97	392
Vascular lesions-5	0.95	0.99	0.97	170	0.99	1.00	0.99	170
Dermatofibroma-6	0.89	0.97	0.93	138	0.99	1.00	1.00	138
Macro avg	0.92	0.94	0.93	5313	0.98	0.98	0.98	5313
Weighted avg	0.92	0.92	0.92	5313	0.98	0.98	0.98	5313

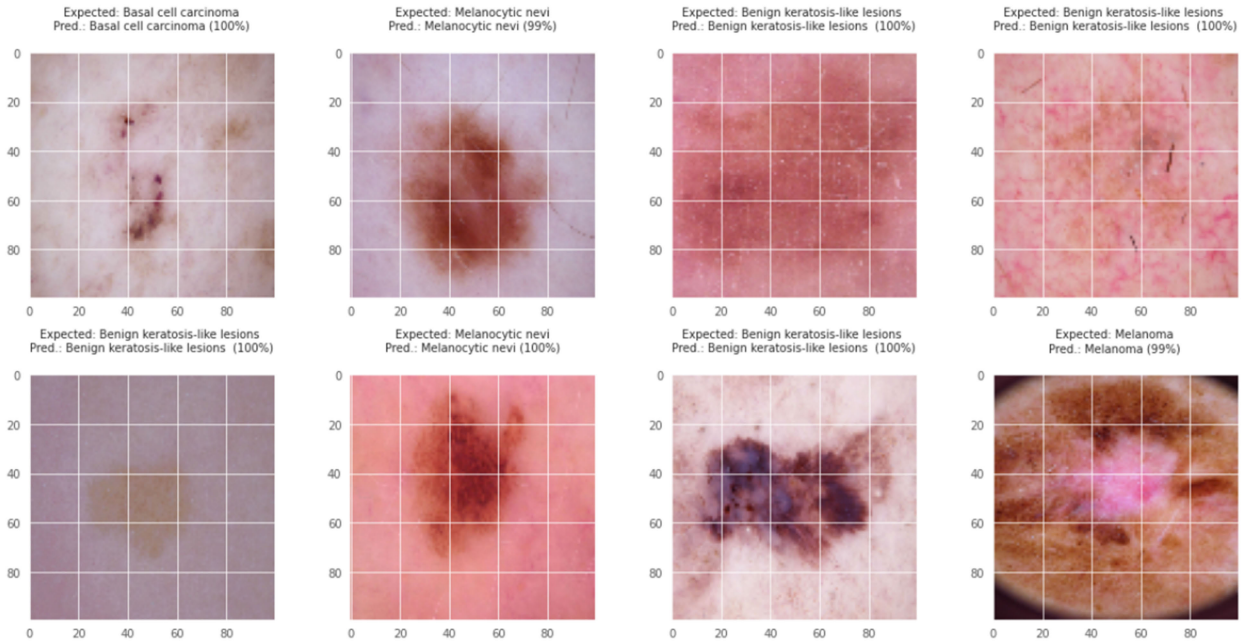


FIGURE 13. Comparative analysis of skin lesion classification results. The image showcases the model's precise predictions for test images, with each labeled to display both the expected and predicted classes.

IncepX-Ensemble models with 100 epochs. This is a crucial graphic to ensure the model gets better and more accurate with each passing epoch as it strives to maximize the goal function. The training accuracy and loss graph plot for the sequential CNN model is shown in the figures above, while the accuracy and loss graph plot for the IncepX-Ensemble model is shown in the graph plot below. The accuracy and loss of a sequential CNN model fluctuate during training. However, in the case of the proposed model, they increase steadily up to the 17th epoch, after which they increase at a moderate rate, and then the growth decreases until the 80th epoch. The loss also declines at roughly the same rates.

Table 7 provides a comprehensive breakdown of each class's precision, recall, and F1-score metrics in both the Sequential CNN model and the proposed IncepX-Ensemble model. These metrics offer insights into the models' classification performance across various skin lesion types. Upon analysis, it is evident that the proposed IncepX-Ensemble model performs better than the Sequential CNN model. For instance, in the class of dermatofibroma, the proposed ensemble model achieves a perfect precision, recall, and F1-score of 1.00, indicating its ability to classify instances of dermatofibroma with precision and sensitivity. Similarly, the class of vascular lesions also attains a perfect classification performance with the proposed ensemble model, achieving precision, recall, and an F1-score of 0.99 or higher. These results highlight the robustness and effectiveness of the proposed model in accurately identifying these specific skin lesion types. Overall, the classification performance of the proposed IncepX-Ensemble model consistently outperforms that of the Sequential CNN model across all classes, reflecting its enhanced capability in discerning subtle distinctions among different skin lesion categories. This suggests that the proposed IncepX-Ensemble model holds promise for improving diagnostic accuracy and reliability in skin lesion classification tasks.

In Fig. 11, the ROC curves show the performance of two models, tried on skin lesion classification: The Sequential

CNN model and the IncepX-Ensemble model. In both the left and right ROC curves, the Sequential CNN model (left ROC curve) exhibits high performance with areas under the curve (AUC) around 0.99 to 1.00 for each class, indicating that the model is remarkably sensitive and specific. As a comparison, the IncepX-Ensemble model (right ROC curve) achieves perfect classification for all classes, with an AUC of 1.00. This concludes that the IncepX-Ensemble model is more accurate and reliable in performing this particular classification task. In summary, both of the models are performing so well, but the IncepX-Ensemble model has perfect separation between true-positive and false-positive rates which will make it more reliable in all classes for skin lesion classification.

Fig. 12 shows a few samples of misclassification or incorrect classification. The predicted result of the model is shown in the figure. An expected (original label of the class) and predicted (recognised by the model) class with an accuracy value is shown at the top of each test image. The percentage indicates how accurately the skin images have been classified. As depicted in the figure, third image from the top left corner, the original is Benign keratosis-like lesions, but the model had misclassified as Melanocytic nevi with a confidence score of 76%.

Fig. 13 compares skin lesion classification results with test images. As shown in the figure, the proposed model correctly predicted most of the test samples. An expected (original label of the class) and predicted (recognised by the model) class with an accuracy value is shown at the top of each test image. For example, from the top left corner, the actual type is benign keratosis-like lesions, and the prediction class is also composed of benign keratosis-like lesions, with a value of 100%. The percentage represents the classification result of the proposed model with a confidence rate of 100%. In this example, the percentage represents the classification result produced by the proposed model, indicating a high level of confidence in the prediction.

V. CONCLUSION

For skin lesion classification using dermoscopy images, an ensemble learning model based on deep learning is proposed. Inaccurate diagnoses and various skin lesions contributed to the patient's death. One of the deadliest illnesses, melanoma, can be fatal if discovered too late. To detect skin lesions early, doctors have developed many methods based on deep learning. As a result of this study, we were able to achieve satisfactory results in diagnosing skin lesions using ensemble techniques. In this experiment, we propose a transfer learning-based ensemble model for the classification of multiclass skin lesions, which provides satisfactory results. In the ensemble, two pre-trained models are combined to create a learning model that can capture rich representations of a variety of images. The proposed research could aid medical professionals in detecting seven different categories of skin ailments earlier. Traditionally, we used data augmentation to balance the dataset classes due to their imbalance. The HAM10000 dataset served as the basis for our experiments. The proposed IncepX-Ensemble model's performance was assessed based on four evaluation metrics in the study. Pre-processing of the images, the splitting of train and test sets, and the use of data enhancement techniques to balance the class data were all carried out prior to the training phase. The findings demonstrate that the improved ensemble transfer learning model outperformed the other suggested transfer learning-based methods, achieving an accuracy of 98%. For the HAM10000 data set, our suggested improved ensembleÂ transfer learning model performs better in classification. Based on a comparison of the proposed model to a recent publication that was conducted on the same data set, IncepX-Ensemble has shown superior accuracy and a lower loss of information. In order to address the problem of data imbalance, we will use GAN-based synthetic data generation in future work to produce more realistic synthetic skin lesion images. Furthermore, we are going to develop a modified version of CycleGAN to generate skin lesion data using stain normalization methods.

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SUBHAJIT CHATTERJEE received the Bachelor of Computer Application (B.C.A.) degree, in 2012, and the Master of Computer Application (M.C.A.) degree, in 2015. He is currently pursuing the Ph.D. degree with the Machine Learning Laboratory, Department of Computer Engineering, Jeju National University, South Korea. He worked related to information technology, software engineering, and customer relationship management of the banking sector for five years. His research interests include artificial intelligence, deep learning, convolutional neural networks, and image processing.



JOON-MIN GIL received the Ph.D. degree in computer science and engineering from Korea University, South Korea, in 2000. He is currently a Professor with the Department of Computer Engineering, Jeju National University (JNU), Jeju, South Korea. Before joining JNU, in September 2023, he was with Daegu Catholic University as a Professor, from March 2006 to August 2023. His recent research interests include cloud computing, big data computing, artificial intelligence, and distributed computing.



YUNG-CHEOL BYUN received the B.S. degree from Jeju National University in 1993, and the M.S. and Ph.D. degrees from Yonsei University in 1995 and 2001, respectively. He worked as a Special Lecturer with SAMSUNG Electronics and SDS from 1998 to 2001. From 2001 to 2003, he was a Senior Researcher with the Electronics and Telecommunications Research Institute (ETRI). He was promoted to join Jeju National University as an Assistant Professor, in 2003, where he is currently a Full Professor with the Computer Engineering Department. His research interests include AI and machine learning, pattern recognition, blockchain and deep learning-based applications, big data and knowledge discovery, time-series data analysis and prediction, image processing, medical image applications, and recommendation systems.